

PharmaWatch

AI-Powered Pharmaceutical Fraud Detection & Consumer Protection Agent

Software Requirements Specification (SRS) — Version 1.0

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1. Introduction

1.1 Purpose of This Document

This Software Requirements Specification (SRS) defines the complete functional requirements, system architecture, agent design, technology stack, and development plan for PharmaWatch — an agentic AI system built to detect pharmaceutical fraud, surface cheaper generic alternatives, and empower Pakistani consumers through collective accountability.

This document guides development during InnoCollab Hackathon 2026 and serves as a reference for judges, mentors, and future contributors.

1.2 Problem Statement

Pakistan's pharmaceutical retail sector is systematically exploitative across three dimensions:

- DRAP sets legally binding Maximum Retail Prices (MRP) for all registered medicines. Pharmacies routinely charge 50%–300% above these official rates with zero consequence.
- Cheaper DRAP-registered generic equivalents exist for almost every branded medicine — yet most patients are never informed of them.
- DRAP publishes Spurious and Substandard Medicine Alerts (counterfeit/unsafe medicines) — yet no consumer tool surfaces these warnings at the point of purchase.
- Filing a formal overcharge complaint requires navigating bureaucratic processes that most citizens cannot manage alone, so violations go unreported.
- Individual complaints are routinely ignored. Collective pattern-based evidence is far harder to dismiss — but no tool exists to build it.

Existing tools such as Dawaai.pk and Pharmapedia Pakistan provide basic medicine price lookup. They do not detect fraud, generate complaints, identify counterfeits, or build community-driven accountability. PharmaWatch is not a price checker. It is a pharmaceutical fraud detection and consumer protection agent.

1.3 Competitive Differentiation

Feature	Dawaai.pk	Pharmapedia PK	PharmaWatch
Medicine price lookup	Yes	Yes	Yes
Generic alternatives	Partial	No	Yes — full active ingredient mapping
Overcharge detection	No	No	Yes — autonomous with evidence
Counterfeit / spurious alerts	No	No	Yes — DRAP alert cross-check
Complaint letter generation	No	No	Yes — pre-filled, evidence-backed PDF
Pharmacy violation history	No	No	Yes — DRAP notices + news search
Crowdsourced fraud heatmap	No	No	Yes — community reports + pattern AI

Feature	Dawaai.pk	Pharmapedia PK	PharmaWatch
Collective case escalation	No	No	Yes — agent-built dossier to DRAP
Agentic multi-step reasoning	No	No	Yes — plan, loop, retry, conclude

1.4 Definitions and Acronyms

Term	Definition
DRAP	Drug Regulatory Authority of Pakistan — official medicine regulator
MRP	Maximum Retail Price — legally binding ceiling price set by DRAP
Generic Medicine	Non-branded drug with the same active ingredient as a branded version
Active Ingredient	The chemically active molecule in a medicine (e.g. Paracetamol in Panadol)
Spurious Medicine	Counterfeit or substandard medicine flagged by DRAP as unsafe
AI Agent	A system that autonomously plans tasks, calls tools, evaluates results, and iterates
Agentic Loop	The plan → execute → evaluate → retry → conclude cycle of true agent behavior
Tool Call	An autonomous API or database query made by the agent without user instruction
Fraud Heatmap	A live map showing pharmacies flagged by community overcharge reports
Collective Dossier	Agent-generated bulk complaint file combining multiple community reports
Firebase	Google's real-time NoSQL cloud database — used for community report storage
SRS	Software Requirements Specification — this document

2. System Overview

2.1 What Makes PharmaWatch a Real AI Agent

Most AI hackathon projects follow a fixed pipeline: user inputs data, LLM processes it, output is displayed. This is not agentic behavior — it is a form submission with a language model attached.

PharmaWatch is architecturally different. Upon receiving a report, the agent:

1. Receives a high-level goal — not step-by-step instructions from the user
2. Independently plans the sequence of investigation steps needed
3. Calls multiple distinct tools in an order it determines dynamically
4. Reads each result and decides the next action based on what it found
5. Retries with reformulated queries when results are insufficient
6. Detects patterns across all historical community reports without user prompting
7. Terminates only when a complete, actionable, evidence-backed output is produced

This plan → tool call → evaluate → loop → conclude cycle is observable in every PharmaWatch interaction and is what judges should recognise as genuine agentic behavior.

2.2 High-Level Agent Workflow

User input

I was charged Rs. 1,200 for Cefum 500mg at City Pharmacy, Saddar, Karachi.

Step	Agent Action	Tool Used	Result
1	Parse: extract medicine, dosage, price, location	Internal NLP	Structured query object
2	DRAP lookup — official MRP and active ingredient	drap_price_lookup	MRP = Rs. 640, Active = Cefuroxime 500mg
3	Evaluate: overcharge Rs. 560 (87.5%). Proceed.	Internal logic	Overcharge flag set
4	Find all DRAP-registered generics with same ingredient	generic_alternatives	Cefim Rs. 390, Zinnat Rs. 520, Xorimax Rs. 410
5	Cross-check DRAP spurious / substandard alerts	spurious_alert_check	No current alert found
6	Search pharmacy violation history	web_search	2 prior DRAP notices found
7	If vague results: reformulate query and retry	web_search (retry)	Improved results or honest null

Step	Agent Action	Tool Used	Result
8	Find nearby pharmacies via location	nearby_pharmacies	4 pharmacies within 2 km
9	Log report anonymously to community database	log_community_report	Report stored; pharmacy now has 13 reports
10	Check if pharmacy crosses pattern threshold	get_pharmacy_reports	13 reports — RED threshold exceeded
11	Generate collective dossier for escalation	generate_collective_dossier	Bulk DRAP complaint with all 13 reports
12	Synthesise all findings into final output	generate_complaint_letter	Full report + PDF complaint letter

2.3 System Components

- Frontend Web Application — React.js user interface with mobile-responsive design
- AI Agent Core — Claude API with tool-calling (claude-sonnet-4-20250514)
- DRAP Local Database — Pre-scraped MRP, generics, and spurious alerts data
- Community Reports Database — Firebase Firestore (anonymous, real-time)
- Pattern Detection Engine — Aggregation queries on Firebase data
- Google Maps Integration — Nearby pharmacy discovery and heatmap overlay
- Web Search Tool — Pharmacy violation history and news lookup
- PDF Generator — Individual complaint letters and collective dossiers

3. Functional Requirements

3.1 Feature 1 — Medicine Price Verification

Purpose

Verify whether the price charged exceeds the DRAP-registered Maximum Retail Price.

Data Source

- DRAP Medicine Registration Portal: drap.gov.pk (publicly accessible, no login required)
- Fields used: Brand Name, Generic Name, Active Ingredient, Strength, MRP (Rs.), Manufacturer
- Implementation: DRAP dataset scraped before the hackathon and stored as local SQLite/JSON — no live dependency on drap.gov.pk during the demo

Agent Behavior

- Normalises medicine name input — handles spelling variants, partial names, brand vs generic
- Looks up MRP from local DRAP dataset
- If medicine not found: agent reformulates query (alternate spelling, generic name) and retries up to 2 times
- Calculates exact overcharge amount (Rs.) and percentage above MRP
- If overcharge confirmed: continues to full investigation pipeline
- If price is within MRP: informs user and still offers generic alternatives and spurious check

Output to User

- Official DRAP MRP with registration number
- Overcharge amount (Rs.) and percentage above MRP
- Manufacturer and DRAP registration reference

3.2 Feature 2 — Generic Alternative Discovery

Purpose

Find all DRAP-registered medicines containing the same active ingredient at a significantly lower price.

How It Works

8. Agent extracts the active ingredient and dosage from the DRAP record (e.g. Paracetamol 500mg for Panadol)
9. Queries the full DRAP registration database for all medicines with matching active ingredient and dosage
10. Filters out the original brand and sorts remaining results by MRP ascending
11. Returns top 3–5 alternatives with brand name, manufacturer, MRP, and registration number

Data Source

- DRAP Registered Medicines database — active ingredient and dosage fields
- All alternatives returned are DRAP-registered — legally sold in Pakistan, verified safe

Why This Is the Most Impactful Feature

Generic alternatives cost 60%–80% less than branded medicines in almost every case, while containing the identical active molecule at the same dosage. A user overcharged for Augmentin at Rs. 1,200 can buy Amoxyclav — same molecule, same DRAP registration — for Rs. 180. This single feature can save Pakistani households thousands of rupees annually.

Output to User

- List of 3–5 generic alternatives: brand name, manufacturer, MRP, DRAP reg. number
- Savings amount vs the overcharged price for each alternative
- Label: same active ingredient, same dosage, DRAP registered

3.3 Feature 3 — Counterfeit & Spurious Medicine Alert

Purpose

Cross-check every queried medicine against DRAP's publicly published Spurious and Substandard Medicine Alerts.

Background

DRAP regularly publishes enforcement alerts listing medicines found to be counterfeit, substandard, or mislabeled. These alerts are public but no consumer application surfaces them at the point of use. A patient buying a flagged medicine has no way of knowing it is dangerous without this feature.

Agent Behavior

- On every medicine query, agent checks local DRAP spurious alerts database
- If a product or batch number matches an active alert: URGENT WARNING displayed immediately above all other results
- Warning includes: alert date, reason (counterfeit / substandard / mislabeled), DRAP notice reference number
- Agent proactively advises: do not consume, report to DRAP, seek verified alternative

Data Source

- DRAP Spurious / Substandard Medicine Alerts — publicly published on drap.gov.pk/enforcement
- Scraped and stored locally; refreshed on each app session startup

Output to User

- Green: No active DRAP alert found for this medicine
- Red urgent banner: DRAP has flagged this product — full alert details displayed

3.4 Feature 4 — Pharmacy Violation History Search

Purpose

Autonomously search for prior DRAP enforcement actions and news reports against the reported pharmacy.

Agent Behavior

- Agent constructs targeted search queries from pharmacy name and area
- Tool call 1: web search — '[pharmacy name] [area] DRAP violation notice'
- Tool call 2: web search — '[pharmacy name] [city] overcharging medicine fine'
- Evaluates results — if vague or irrelevant, reformulates with alternate terms and retries once
- Also checks community database: how many prior PharmaWatch reports exist for this pharmacy

Data Sources

- DRAP Enforcement Notices — publicly published on drap.gov.pk
- News archives — Dawn, Geo, Express Tribune (accessible via web search)
- PharmaWatch Firebase — prior community reports for this pharmacy

Output to User

- Prior violations found: list with dates, nature of violation, source link
- No violations found: stated clearly — agent never fabricates results
- Community report count: total PharmaWatch reports for this pharmacy with overcharge summary

3.5 Feature 5 — DRAP Complaint Letter Generation

Purpose

Automatically generate a formal, pre-filled complaint letter addressed to DRAP with all evidence embedded.

Letter Contents

- Complainant reference (anonymous or named — user's choice)
- Pharmacy name, area, and city
- Medicine name, strength, and DRAP registration number
- Official MRP with DRAP source citation
- Amount charged, overcharge in Rs., and percentage above MRP
- Date of incident
- Prior violations found by agent, if any, with source references
- Formal request for investigation under the DRAP Act 2012
- Addressed to: Drug Regulatory Authority of Pakistan, Islamabad

Technical Implementation

- Agent generates letter content using structured evidence data from all prior tool calls
- PDF export via jsPDF library — client-side, no server required
- One-click download as PDF, one-click copy as plain text

Why This Matters

The barrier to filing a complaint in Pakistan is not willingness — it is capability. Most citizens do not know how to write a formal complaint, what legal language to use, or where to send it. PharmaWatch removes every single part of that barrier. The agent does the legal drafting; the user downloads and submits.

3.6 Feature 6 — Nearby Pharmacy Finder

Purpose

Surface nearby verified pharmacies so the user can immediately find a safe, legitimate alternative.

Agent Behavior

- Agent calls Google Maps Places API with user's location or entered area name
- Returns 4–5 nearest pharmacies with name, address, distance, and Google rating
- Overlays community fraud data: if a returned pharmacy is flagged on the heatmap, it is marked in orange or red

Honest Scope Statement

Real-time stock availability is not included in this version. No public API exists for pharmacy inventory in Pakistan. Judges will be informed of this transparently. The feature provides verified locations — stock confirmation is a Phase 2 integration requiring direct pharmacy partnerships.

Output to User

- Map view with pins for nearest pharmacies
- List view with name, address, and distance
- Fraud status overlay from heatmap: clean / watch / suspicious / flagged

3.7 Feature 7 — Crowdsourced Fraud Heatmap

Purpose

Every anonymous user report is stored and analyzed collectively. The agent detects patterns, identifies serial offenders, and builds cases no individual could build alone.

How Reports Are Collected

Every time a user submits an overcharge report through PharmaWatch, the agent silently and anonymously logs the following — no opt-in required, no extra steps for the user:

- Pharmacy name and area
- Medicine overcharged and by how much
- City and timestamp
- No name, phone number, CNIC, or any personal identifier is ever stored

Pattern Detection Thresholds

Reports in 30 Days	Classification	Map Marker	Agent Action
1 – 3 reports	Watch	Yellow dot	Monitor — no public action
4 – 9 reports	Suspicious	Orange dot	Visible to all users on heatmap

Reports in 30 Days	Classification	Map Marker	Agent Action
10+ reports	Confirmed	Red dot	Trigger collective escalation dossier

Collective Case Escalation (10+ reports)

When a pharmacy crosses the threshold, the agent autonomously — without any user requesting it:

12. Aggregates all reports for that pharmacy into a structured evidence dataset
13. Cross-references each reported medicine against DRAP MRP to verify every overcharge claim
14. Builds a collective dossier: pharmacy profile, total incidents, medicines involved, average overcharge %, full timeline
15. Generates a bulk DRAP complaint letter citing all verified incidents with anonymous contributor count
16. Marks pharmacy as RED on the public heatmap visible to all users in real time

One complaint is easy to ignore. A formally documented dossier citing 12 verified incidents, submitted to DRAP, is a serious legal matter. PharmaWatch makes this possible with zero additional effort from any individual user.

Heatmap Interface

- Pakistan map loads as the application homepage — visible before any search is made
- Colored dots mark all pharmacies with community reports, clustered by city
- Clicking a dot shows: total reports, medicines flagged, average overcharge %, date of most recent report
- Download Collective Complaint button appears on all RED pharmacies
- New submissions update the map in real time via Firebase's live sync

Privacy and Anonymization

- No PII (name, phone, CNIC, email, IP address) is ever stored
- Data stored per report: pharmacy name, area, city, medicine, price discrepancy, timestamp
- Session ID stored as a one-way hash for rate-limiting only — cannot be reversed to identify a user

Technical Implementation

- Firebase Firestore — free tier, real-time, sufficient for Pakistan-scale demo load
- Aggregation query: COUNT reports WHERE pharmacyName = X AND timestamp > 30 days ago
- Google Maps JS API — custom colored markers overlaid on standard map tiles

4. Non-Functional Requirements

Category	Requirement	Target / Notes
Performance	Full agent pipeline completion time	Under 15 seconds per query
Reliability	DRAP database availability during demo	100% — local dataset, zero external dependency
Scalability	Community report writes	Real-time Firebase handles concurrent submissions
Usability	Device compatibility	Fully functional on mobile browsers (responsive design)
Language	Interface language	English for hackathon; Urdu planned for Phase 2
Privacy	Personal data in community reports	Zero PII stored — anonymous by design
Security	Firebase read/write rules	Read: public (heatmap). Write: rate-limited per session.
Transparency	Agent reasoning visibility	Collapsible tool-call trace panel shown to user
Accessibility	Complaint letter formats	PDF download and plain text copy both provided
Honesty	Data limitations disclosed to judges	Real-time stock: not available — stated clearly in UI

5. Technology Stack

5.1 Frontend

Technology	Purpose	Justification
React.js	Core UI framework	Fast component-based dev; both team members know JavaScript
Tailwind CSS	Styling	Rapid UI design without custom CSS from scratch
Google Maps JavaScript API	Interactive heatmap + nearby pharmacies	Most complete pharmacy location data for Pakistan; custom marker support
jsPDF	Complaint PDF generation	Client-side PDF — no additional server required
Axios	HTTP requests to agent server	Clean async API call handling

5.2 AI Agent Backend

Technology	Purpose	Justification
Claude API (claude-sonnet-4-20250514)	Core agent reasoning and tool orchestration	Best-in-class tool-calling, multi-step planning, reliable structured output
Node.js + Express	Agent API server	Lightweight, fast to set up, JavaScript end-to-end
Anthropic Tool Use API	True agentic behavior	Agent decides which tools to call and in what order — not hardcoded sequence

5.3 Data Layer

Technology	Purpose	Justification
Firebase Firestore	Community reports + real-time heatmap sync	Free tier, real-time updates, Google ecosystem — aligns with sponsor
SQLite / JSON	Local DRAP medicine database	Static data; fast local lookup; zero external dependency during demo
Python scraper	Pre-hackathon DRAP data collection	One-time run before hackathon to build MRP + generics + spurious alerts DB
Google Maps Places	Nearby pharmacy discovery	Most accurate pharmacy location data available for Pakistani cities

5.4 External APIs & Data Sources

Source	Data	Access	Cost
drap.gov.pk	MRP list, generic registrations, spurious alerts	Public website — scraped pre-hackathon	Free
Google Maps Places API	Nearby pharmacy locations	REST API with key	Free tier
Anthropic Claude API	Agent reasoning and text generation	REST API with key	Pay-per-token
Firebase Firestore	Community report storage and real-time sync	Google SDK	Free tier
Web Search (Claude)	Pharmacy violation history and news	Built into Claude API	Included

6. Data Architecture

6.1 DRAP Local Database Schema

Built from DRAP public data before the hackathon. Stored as SQLite or structured JSON — no live internet dependency during the demo.

Table	Key Fields	Source
medicines	brand_name, generic_name, active_ingredient, strength, mrp, manufacturer, reg_number	DRAP MRP list
generics_map	active_ingredient, strength, brand_name, mrp, manufacturer, reg_number	DRAP registered products
spurious_alerts	medicine_name, alert_date, reason, batch_number, drap_notice_ref	DRAP spurious alerts page

6.2 Firebase Community Reports Schema

One Firestore document per report. No PII ever stored.

Field	Type	Description
pharmacyName	String	Pharmacy name as entered by user
area	String	Neighborhood (e.g. Gulshan, PECHS, Saddar)
city	String	City (e.g. Karachi, Lahore, Islamabad)
medicine	String	Medicine name reported overcharged
officialMRP	Number	DRAP MRP at time of report
chargedPrice	Number	Price the user was actually charged
overchargeAmt	Number	Difference: chargedPrice minus officialMRP
overchargePct	Number	Percentage above official MRP
timestamp	Timestamp	Date and time of report submission
sessionId	String	One-way hash for rate-limiting only — cannot identify user

7. Agent Design — Tool Definitions

The following nine tools are registered with the Claude agent. The agent autonomously decides which tools to call, in what order, and whether to retry based on results received.

Tool Name	Description	Key Inputs	Key Outputs
drap_price_lookup	Fetch official MRP and active ingredient	medicine_name, strength	mrp, active_ingredient, manufacturer, reg_number
generic_alternatives	Find DRAP-registered generics with same active ingredient	active_ingredient, strength	Array of {brand, mrp, manufacturer, reg_number}
spurious_alert_check	Check DRAP spurious / substandard alerts	medicine_name	alert: true/false, alert_details if flagged
web_search	Search pharmacy violation history in DRAP notices and news	query string	Snippets with source URLs
nearby_pharmacies	Find nearest pharmacies to a given location	area, city	Array of {name, address, distance, rating}
log_community_report	Store overcharge report anonymously to Firebase	pharmacyName, area, city, medicine, prices	Confirmation + updated report count
get_pharmacy_reports	Retrieve aggregated community reports for a pharmacy	pharmacyName, area	Report count, medicines, avg overcharge, status
generate_complaint_letter	Generate formal DRAP complaint letter with all evidence	All investigation findings	Formatted letter text for PDF export
generate_collective_dossier	Build bulk case file from all community reports	pharmacyName, area	Dossier with all reports + bulk complaint letter

7.1 Retry Logic

The following retry behaviors distinguish PharmaWatch from a fixed pipeline:

- drap_price_lookup returns no result: agent reformulates medicine name (generic name, alternate spelling) and retries up to 2 times
- web_search returns vague or irrelevant results: agent reformulates query with alternate terms and retries once
- nearby_pharmacies returns empty set: agent broadens the search to city level
- All retries exhausted: agent honestly reports 'not found' — data is never fabricated

8. Hackathon Development Plan

8.1 Team Split

Member	Role	Key Deliverables
Member 1	Frontend	React UI, Google Maps heatmap, PDF export, UX, mobile responsiveness
Member 2	Backend / Agent	Claude agent server, all 9 tool implementations, DRAP scraper, Firebase integration

8.2 Day-by-Day Build Plan

Phase	Task	Owner	Est. Time
Pre-hackathon	Scrape DRAP MRP, generics, and spurious alerts into SQLite/JSON	Member 2	2–3 hrs
Pre-hackathon	Set up Firebase project + Firestore schema + env keys	Member 2	30 min
Pre-hackathon	Create React + Tailwind app scaffold	Member 1	30 min
Day 1 – Morning	Build Claude agent with all 9 tool definitions and basic loop	Member 2	4 hrs
Day 1 – Morning	Build medicine search input + DRAP result display UI	Member 1	4 hrs
Day 1 – Afternoon	Integrate generic alternatives + spurious alert display	Both	3 hrs
Day 1 – Afternoon	Build complaint letter generation + jsPDF export	Both	2 hrs
Day 1 – Evening	Build community report logging + Firebase write integration	Member 2	2 hrs
Day 1 – Evening	Build Google Maps heatmap with colored markers per threshold	Member 1	2 hrs
Day 2 – Morning	Build pattern detection logic + collective dossier generation	Member 2	3 hrs
Day 2 – Morning	Polish heatmap UI — pharmacy popup, report counter, dossier button	Member 1	3 hrs
Day 2 – Afternoon	End-to-end testing, bug fixes, demo flow rehearsal	Both	3 hrs
Day 2 – Evening	Pitch slides and demo script preparation	Both	2 hrs

8.3 Demo Script for Judges

Designed to show maximum agentic behavior in approximately 3 minutes:

17. App opens. Pakistan heatmap is visible. One red dot in Saddar, Karachi. Click it: '13 verified overcharge reports. Medicines: Augmentin, Ceftum, Brufen. Average overcharge: 164%.'
18. User submits: 'Charged Rs. 1,200 for Ceftum 500mg at City Pharmacy, Saddar.'
19. Agent trace panel opens — judges watch all tool calls fire in real time: DRAP lookup, generic search, spurious check, web search (with reformulated retry), community lookup, nearby pharmacies.
20. Results displayed: overcharge Rs. 560 confirmed (87%). Three generic alternatives shown, cheapest saves Rs. 820. No spurious alert. Two prior DRAP notices found. Four nearby pharmacies marked on map.
21. Community report logged. Red dot on Saddar pharmacy updates to 14 reports in real time.
22. Judge clicks 'Download Collective Complaint' — PDF dossier with all 14 reports generates instantly and downloads.

9. Risk Register

Risk	Likelihood	Impact	Mitigation
DRAP website structure changes before hackathon	Low	High	Scrape all data pre-hackathon and store locally. Demo has zero live dependency on drap.gov.pk.
Claude API rate limits during live demo	Medium	High	Cache all DRAP lookups locally. Only synthesis uses LLM. Keep demo queries simple and pre-tested.
Google Maps API quota exceeded during demo	Low	Medium	Prepare fallback with pre-loaded pharmacy data displayed as static list if Maps API fails.
Firebase real-time sync lag during demo	Low	Medium	Pre-seed heatmap with realistic data before demo so map is visually impressive regardless.
Agent hallucinates DRAP prices	Medium	High	All DRAP lookups use local database. LLM never generates medicine prices — only synthesises text from verified data.
Scope too ambitious for 2 people	Medium	High	Features 1–5 are the MVP. Features 6–7 are bonus. Ship core features first, add heatmap if time allows.

10. Pitch Framework

10.1 The One-Line Pitch

Pitch

PharmaWatch is Pakistan's first agentic AI that detects pharmacy fraud, finds you cheaper safe alternatives, files your complaint, and protects your entire neighborhood — powered entirely by DRAP's own public data.

10.2 The Differentiation Line

vs. Competitors

Dawaai.pk tells you what a medicine should cost. PharmaWatch proves you were defrauded, finds you a safer cheaper alternative, generates your legal complaint, and alerts your community so it never happens to them.

10.3 The Emotional Hook

Right now in Pakistan, a person gets overcharged at a pharmacy. They have two options: accept it because they don't know better, or try to file a complaint alone — with no evidence, no legal knowledge, against a licensed business. They almost always accept it.

PharmaWatch gives them a third option. Their report joins thirteen others. Together they become an unanswerable case. The agent writes the legal dossier. One tap submits it to DRAP.

That is not a price checker. That is justice infrastructure.

10.4 Sponsor Alignment

Sponsor	Alignment with PharmaWatch
Google for Developers	Google Maps API powers the heatmap and nearby pharmacy feature. Firebase provides real-time community data sync. Architecture is compatible with Gemini API as an alternative model.
Telenor 4G	Designed for mobile use on 4G networks. Low-bandwidth responsive UI. Urdu support in Phase 2 directly targets Telenor's rural and low-literacy subscriber base. SMS alert integration is a natural Phase 2 feature via Telenor's network.
Ministry of IT & Telecom (MoITT)	Directly advances the Digital Pakistan vision. Strengthens DRAP enforcement capacity using citizen-generated data. Aligns with e-governance, consumer protection, and public health mandates.
Innovista	Clear post-hackathon startup trajectory. B2B version available for NGOs, consumer protection bodies, and government regulators. Model replicable in any country with a public drug pricing authority.

Appendix A — Public Data Sources

Source	URL	Data Available
DRAP MRP List	drap.gov.pk/medicine-prices	Official maximum retail prices for all registered medicines in Pakistan
DRAP Registered Products	drap.gov.pk/registered-products	Full registration database with active ingredients, dosages, and manufacturers
DRAP Spurious Alerts	drap.gov.pk/spurious-medicines	Counterfeit and substandard medicine enforcement warnings
DRAP Enforcement Notices	drap.gov.pk/enforcement	Pharmacy violation notices, fines, and suspension orders
Google Maps Places API	developers.google.com/maps/places	Pharmacy locations, addresses, and ratings across Pakistan
Firebase Firestore	firebase.google.com/products/firestore	Real-time NoSQL database for community report storage
Pakistan Bureau of Stats	pbs.gov.pk	District-level demographic and economic context data

Appendix B — Feature Priority Matrix

Feature	Priority	Hackathon MVP?	Est. Build Time
Medicine Price Verification	P0 — Critical	Yes	3–4 hours
Generic Alternative Discovery	P0 — Critical	Yes	3–4 hours
Counterfeit / Spurious Alert	P0 — Critical	Yes	2 hours
DRAP Complaint Letter Generation	P1 — High	Yes	2–3 hours
Pharmacy Violation History Search	P1 — High	Yes	2 hours
Nearby Pharmacy Finder	P1 — High	Yes	2 hours
Crowdsourced Fraud Heatmap	P1 — High	Yes	4–5 hours
Collective Case Escalation Dossier	P2 — Medium	Stretch goal	3 hours
Urdu Language Support	P3 — Low	No — Phase 2	Post-hackathon
SMS Alerts via Telenor	P3 — Low	No — Phase 2	Post-hackathon
Native Mobile App	P3 — Low	No — Phase 2	Post-hackathon

PharmaWatch — InnoCollab Hackathon 2026 — SRS v1.0

Built with purpose. Powered by public data. Protecting Pakistan one report at a time.